MIPT & YSDA · AUTUMN 2026

Deep Generative Models

Lecture 11

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Recap of Previous Lecture

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

Variance Exploding SDE (NCSN)

This process matches the NCSN noise levels.

$$d\mathbf{x} = \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}$$

$$\mathbf{f}(\mathbf{x}, t) = 0, \quad g(t) = \sqrt{\frac{d[\sigma^2(t)]}{dt}}$$

$\sigma(t)$ is a monotonically increasing function.

Variance Preserving SDE (DDPM)

This is the small-step limit of the DDPM forward chain.

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}(t)dt + \sqrt{\beta(t)} \cdot d\mathbf{w}$$

$$\mathbf{f}(\mathbf{x}, t) = -\frac{1}{2}\beta(t)\mathbf{x}(t), \quad g(t) = \sqrt{\beta(t)}$$

Variance is preserved as long as $\mathbf{x}(0)$ has unit variance.

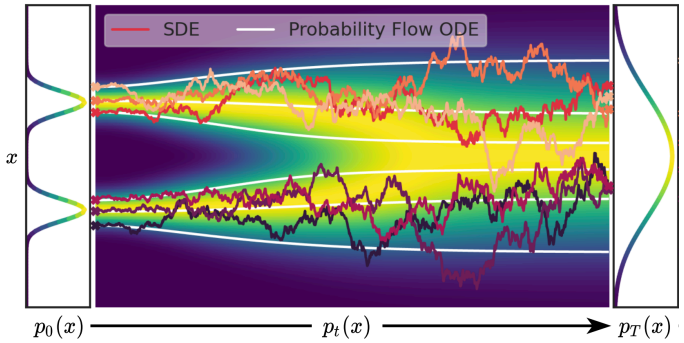
Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020

Recap of Previous Lecture

Theorem

Suppose the SDE $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ induces the probability path $p_t(\mathbf{x})$ with $p_t(\mathbf{x}) > 0$. Then, there exists an ODE with the same probability path $p_t(\mathbf{x})$, given by

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt = \left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2} g^2(t) \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) dt$$



Log-likelihood

The flow is invertible. Follow the same ODE from data to noise to evaluate the model density.

$$\log p_{\boldsymbol{\theta}}(\mathbf{x}(0)) = \log p_T(\mathbf{x}(T)) + \int_0^T \operatorname{tr}\left(\frac{\partial \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)} \right) dt.$$

Recap of Previous Lecture

ODE · One predecessor

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)\, dt$$

Over a short forward step h,

$$\mathbf{x}(t+h) \approx \mathbf{x}(t) + h\, \mathbf{v}(\mathbf{x}(t), t).$$

In reverse time $\tau = 1 - t$,

$$d\mathbf{x} = -\mathbf{v}(\mathbf{x}, 1-\tau)\, d\tau.$$

Theorem

There exists a reverse SDE for $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ (assume $p_t(\mathbf{x}) > 0$), given by:

$$d\mathbf{x} = \left(\mathbf{f}(\mathbf{x}, t) - g^2(t) \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})\right) dt + g(t) d\bar{\mathbf{w}}$$

where $dt < 0$ and $\bar{\mathbf{w}}(t)$ is a reverse-time Wiener process.

Anderson B. D. O. Reverse-time diffusion equation models, 1982
Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020

Recap of Previous Lecture

Proof Sketch

- Convert the initial SDE to a probability flow ODE.
- Reverse the probability flow ODE.
- Convert the reversed probability flow ODE back to an SDE.

Recap of Previous Lecture

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

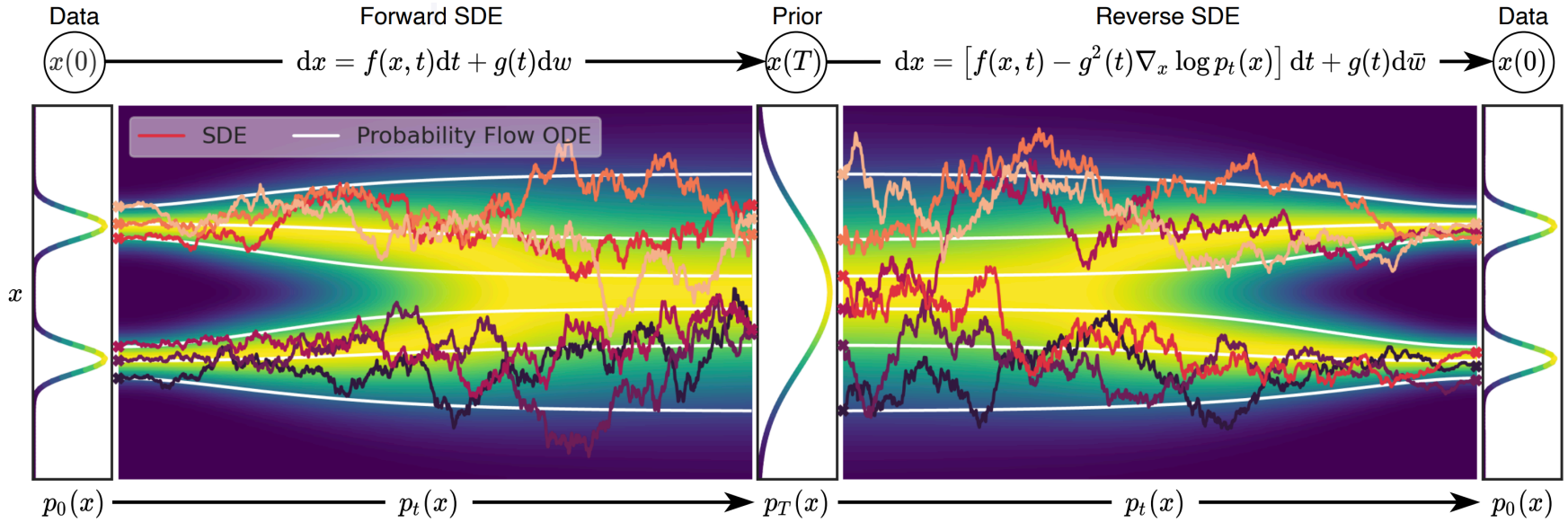
$$\mathbf{v}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\mathbf{s}(\mathbf{x}, t), \quad \text{where} \quad \mathbf{s}(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log p_t(\mathbf{x})$$

Probability Flow ODE

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)\, dt.$$

Reverse SDE

$$d\mathbf{x} = \left(\mathbf{v}(\mathbf{x}, t) - \tfrac{1}{2}g^2(t)\mathbf{s}(\mathbf{x}, t)\right) dt + g(t)\, d\bar{\mathbf{w}}.$$



Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020

Recap of Previous Lecture

Discrete-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)}\mathbb{E}_{t\sim U\{1,T\}}\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}\left\|\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t}\log q(\mathbf{x}_t|\mathbf{x}_0)\right\|_2^2$$

Use unit score-loss weights here.

Continuous-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))}\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{q(\mathbf{x}(t)|\mathbf{x}(0))}\left\|\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}(t),t) - \nabla_{\mathbf{x}(t)}\log q(\mathbf{x}(t)\mid\mathbf{x}(0))\right\|_2^2$$

In continuous time, the conditional law is still Gaussian. Write $\boldsymbol{\mu} = \boldsymbol{\mu}(\mathbf{x}(0), t)$ for its mean.

Conditional density

$$q(\mathbf{x}(t)\mid\mathbf{x}(0)) = \mathcal{N}\big(\boldsymbol{\mu},\, \sigma^2(t)\mathbf{I}\big)$$

Score target

$$\nabla_{\mathbf{x}(t)}\log q(\mathbf{x}(t)\mid\mathbf{x}(0)) = \frac{\boldsymbol{\mu} - \mathbf{x}(t)}{\sigma^2(t)}$$

Recap of Previous Lecture

With affine drift and additive Gaussian noise, the transition law is Gaussian.

$$d\mathbf{x} = f(t)\mathbf{x}\,dt + g(t)\,d\mathbf{w} \quad \Longrightarrow \quad q(\mathbf{x}(t) \mid \mathbf{x}(0)) = \mathcal{N}\left(\boldsymbol{\mu}(\mathbf{x}(0), t),\, \sigma^2(t)\mathbf{I}\right).$$

Theorem

$$\frac{d\boldsymbol{\mu}(\mathbf{x}(0), t)}{dt} = f(t)\boldsymbol{\mu}(\mathbf{x}(0), t), \qquad \frac{d\sigma^2(t)}{dt} = 2f(t)\sigma^2(t) + g^2(t).$$

Fix $\mathbf{x}(0)$ and write $v(t)$ for the conditional variance per coordinate. Then $\boldsymbol{\mu}(0) = \mathbf{x}(0)$ and $v(0) = 0$.

Recap of Previous Lecture

Training

1. Sample $\mathbf{x}(0) \sim p_{\text{data}}(\mathbf{x})$, $t \sim U[0, 1]$.
2. Sample $\mathbf{x}(t) \sim q(\mathbf{x}(t)|\mathbf{x}(0))$.
3. Compute loss $\mathcal{L} = \left\|\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}(t), t) - \nabla_{\mathbf{x}(t)} \log q(\mathbf{x}(t)|\mathbf{x}(0))\right\|_2^2$.

NCSN

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\mathbf{x}(0), \left[\sigma^2(t) - \sigma^2(0)\right] \cdot \mathbf{I}\right)$$

DDPM

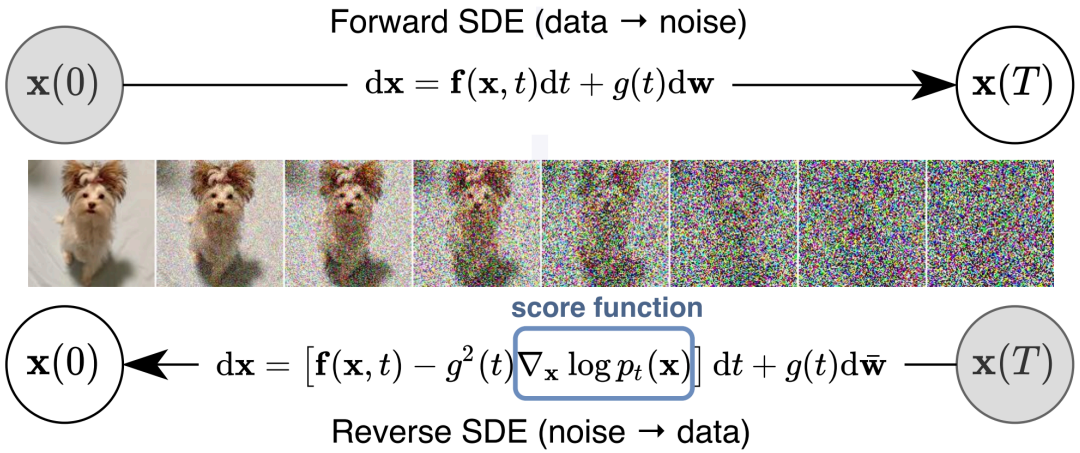
$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\mathbf{x}(0) e^{-\frac{1}{2}\int_0^t \beta(s)ds}, \left(1 - e^{-\int_0^t \beta(s)ds}\right) \cdot \mathbf{I}\right)$$

Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020

Recap of Previous Lecture

Sampling

1. For the VP-SDE, sample $\mathbf{x}(1) \sim \mathcal{N}(0, \mathbf{I})$ from the approximate terminal distribution.
2. Solve the reverse SDE from $T = 1$ to $t = 0$ using `SDESolve`.



- An Euler–Maruyama step of the reverse SDE adds fresh Gaussian noise.
- An Euler step of the probability flow ODE is deterministic given the current state.

Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020

Outline

01 Denoiser Parameterization and Sampling

02 Flow Matching (FM)

03 Conditional Flow Matching (CFM)

CFM Objective
Conditional Probability Paths

04 One-Sided Conditioning

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From Time to Noise Level

Use the familiar additive-noise form: $\mathbf{x} = \mathbf{x}_0 + \sigma\boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$.

$$\mathbf{f}(\mathbf{x}, t) = 0, \qquad g^2(t) = \frac{d\sigma^2(t)}{dt} = 2\sigma(t)\frac{d\sigma(t)}{dt}.$$

Rewrite the probability flow ODE using σ

$$\begin{aligned}\frac{d\mathbf{x}(t)}{dt} &= \mathbf{f}(\mathbf{x}(t), t) - \frac{1}{2}g^2(t)\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}(t), \sigma(t)) \\ &= -\sigma(t)\frac{d\sigma(t)}{dt}\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}(t), \sigma(t)).\end{aligned}$$

$$\frac{d\mathbf{x}}{d\sigma} = \left(\frac{d\sigma(t)}{dt}\right)^{-1}\frac{d\mathbf{x}(t)}{dt} = -\sigma\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}, \sigma).$$

Choose a denoiser using Tweedie's formula

$$\mathbb{E}[\mathbf{x}_0 \mid \mathbf{x}] = \mathbf{x} + \sigma^2\nabla_{\mathbf{x}}\log p_\sigma(\mathbf{x}).$$

$$D_{\boldsymbol{\theta}}(\mathbf{x}, \sigma) \approx \mathbb{E}[\mathbf{x}_0 \mid \mathbf{x}], \qquad \frac{d\mathbf{x}}{d\sigma} = -\sigma\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}, \sigma) = \frac{\mathbf{x} - D_{\boldsymbol{\theta}}(\mathbf{x}, \sigma)}{\sigma}.$$

Decreasing σ moves towards the denoiser prediction. **How do we train this denoiser?**

Train the Denoiser

Use the existing score-training objective. For a noisy-clean pair $(\mathbf{x}, \mathbf{x}_0)$ with $q(\mathbf{x} \mid \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_0, \sigma^2 \mathbf{I})$:

$$\mathcal{L} = \left\| \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}, \sigma) - \nabla_{\mathbf{x}} \log q(\mathbf{x} \mid \mathbf{x}_0) \right\|_2^2 .$$

Compute the conditional Gaussian target

$$\nabla_{\mathbf{x}} \log q(\mathbf{x} \mid \mathbf{x}_0) = -\frac{\mathbf{x} - \mathbf{x}_0}{\sigma^2}.$$

Substitute the denoiser representation

$$\begin{aligned} \mathcal{L} &= \left\| \frac{D_{\boldsymbol{\theta}}(\mathbf{x}, \sigma) - \mathbf{x}}{\sigma^2} + \frac{\mathbf{x} - \mathbf{x}_0}{\sigma^2} \right\|_2^2 \\ &= \frac{1}{\sigma^4} \left\| D_{\boldsymbol{\theta}}(\mathbf{x}, \sigma) - \mathbf{x}_0 \right\|_2^2 . \end{aligned}$$

Preserving the score objective requires the factor $1/\sigma^4$.

Plain denoiser MSE would reweight noise levels. The trained denoiser supplies the sampling field.

What Does the Optimal Denoiser Predict?

Use uniform empirical data $\{\mathbf{x}_1, \ldots, \mathbf{x}_N\}$. For fixed $\mathbf{x}$ and $\sigma > 0$, let $a_i = \mathcal{N}(\mathbf{x}; \mathbf{x}_i, \sigma^2 \mathbf{I})$ be the likelihood of the noisy observation.

Minimize the conditional MSE at $\mathbf{x}$

$$D^*(\mathbf{x}, \sigma) = \arg\min_{\mathbf{y}} \sum_{i=1}^{N} a_i \|\mathbf{y} - \mathbf{x}_i\|_2^2 \quad \Longrightarrow \quad 2 \sum_{i=1}^{N} a_i (\mathbf{y} - \mathbf{x}_i) = 0.$$

$$D^*(\mathbf{x}, \sigma) = \frac{\sum_{i=1}^{N} \exp\left(-\frac{\|\mathbf{x}-\mathbf{x}_i\|_2^2}{2\sigma^2}\right) \mathbf{x}_i}{\sum_{j=1}^{N} \exp\left(-\frac{\|\mathbf{x}-\mathbf{x}_j\|_2^2}{2\sigma^2}\right)} = \mathbb{E}[\mathbf{x}_0 \mid \mathbf{x}].$$

Set the gradient to zero: the optimum is a **posterior-weighted average**.

Why train a network?

- The exact formula scans **all N training examples at every sampling step**.
- It describes the empirical distribution; exact sampling only reproduces training examples.

Generate Samples with a Denoiser

Use the trained denoiser in the ODE solver. Start from strong noise and decrease σ.

Sampling (Euler)

1. Choose $\sigma_0 > \sigma_1 > \cdots > \sigma_N = 0$ and sample $\mathbf{x}^{(0)} \sim \mathcal{N}(0, \sigma_0^2 \mathbf{I})$.
2. For $i = 0, \ldots, N-1$, evaluate $D_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}, \sigma_i)$ and update:

$$\mathbf{x}^{(i+1)} = \mathbf{x}^{(i)} + (\sigma_{i+1} - \sigma_i) \frac{\mathbf{x}^{(i)} - D_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}, \sigma_i)}{\sigma_i}.$$

3. Return $\mathbf{x}^{(N)}$.

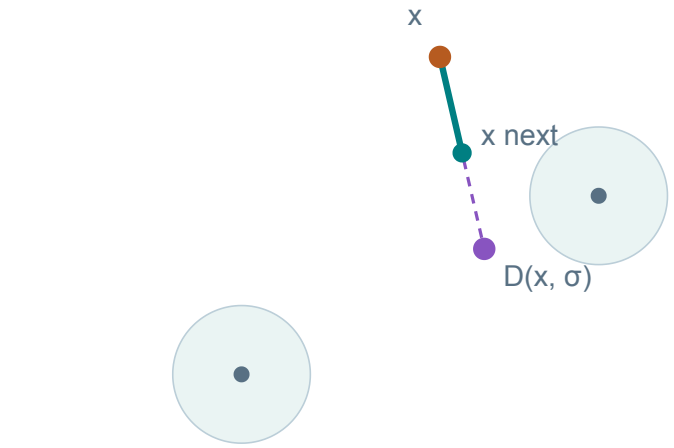
We now have an explicit Euler sampler in noise coordinates.
How does this relate to the deterministic DDIM update?

One Denoiser Prediction, One Euler Step

Decrease the noise level, then recompute the denoiser at the new state.

$$\mathbf{x}_{\text{next}} = r\mathbf{x} + (1-r)D_{\boldsymbol{\theta}}(\mathbf{x}, \sigma), \qquad r = \sigma_{\text{next}}/\sigma$$

One Euler step: σ = 2.5 → 1.25. The next state lies halfway to the current denoiser prediction.



Shaded: clean-data modes

Noise level

2.50 → 1.25

Move partway towards the current prediction.

After each step, recompute the denoiser at the new point and noise level.

For r = 0, the final step returns the current prediction.

An Euler sampling step interpolates towards the denoiser prediction. DDIM will give the same update in noise coordinates.

Exact conditional mean for a smooth two-Gaussian toy. The conditional mean can lie between modes.

DDIM Is Euler in Noise Coordinates

Recall deterministic DDIM ($s < t$):

$$\hat{\boldsymbol{\epsilon}} = \boldsymbol{\epsilon}_{\boldsymbol{\theta},t}(\mathbf{x}_t), \qquad \hat{\mathbf{x}}_0 = \frac{\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\,\hat{\boldsymbol{\epsilon}}}{\sqrt{\bar{\alpha}_t}},$$

$$\mathbf{x}_s = \sqrt{\bar{\alpha}_s}\,\hat{\mathbf{x}}_0 + \sqrt{1-\bar{\alpha}_s}\,\hat{\boldsymbol{\epsilon}}.$$

Recall the DDPM / NCSN rescaling ($\sigma_s < \sigma_t$)

$$\tilde{\mathbf{x}}_u = \frac{\mathbf{x}_u}{\sqrt{\bar{\alpha}_u}}, \qquad \sigma_u = \sqrt{\frac{1-\bar{\alpha}_u}{\bar{\alpha}_u}}, \qquad u \in \{s, t\}.$$

$$D_{\boldsymbol{\theta}}(\tilde{\mathbf{x}}_t, \sigma_t) = \hat{\mathbf{x}}_0 = \tilde{\mathbf{x}}_t - \sigma_t \hat{\boldsymbol{\epsilon}}.$$

Rewrite DDIM as the denoiser Euler step

$$\begin{aligned}\tilde{\mathbf{x}}_s &= \hat{\mathbf{x}}_0 + \sigma_s \hat{\boldsymbol{\epsilon}} = \tilde{\mathbf{x}}_t + (\sigma_s - \sigma_t)\hat{\boldsymbol{\epsilon}} \\ &= \tilde{\mathbf{x}}_t + (\sigma_s - \sigma_t)\frac{\tilde{\mathbf{x}}_t - D_{\boldsymbol{\theta}}(\tilde{\mathbf{x}}_t, \sigma_t)}{\sigma_t}.\end{aligned}$$

DDIM implements this Euler step using the existing noise predictor.

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01 Denoiser Parameterization and Sampling

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Continuous-Time Normalizing Flows

Now learn the velocity field itself. Recall $\mathbf{x}_t = \mathbf{x}(t)$ for $t \in [0, 1]$:

- $\mathbf{x}_0 \sim p_0(\mathbf{x})$, where $p_0(\mathbf{x})$ is a base distribution (mostly $\mathcal{N}(0, \mathbf{I})$);
- $\mathbf{x}_1 \sim p_1(\mathbf{x})$, where $p_1(\mathbf{x})$ is the true data distribution $p_{\text{data}}(\mathbf{x})$.

Time convention: unlike the diffusion sections, here $t = 0$ denotes noise and $t = 1$ denotes data. Sampling runs forward in time.

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, t), \quad \text{with initial condition } \mathbf{x}(0) = \mathbf{x}_0.$$

The velocity $\mathbf{v}$ determines a flow $\boldsymbol{\psi}_t$: $\mathbf{x}_t = \boldsymbol{\psi}_t(\mathbf{x}_0)$.

KFP Theorem (Continuity Equation)

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}\left(\mathbf{v}(\mathbf{x}, t) p_t(\mathbf{x})\right) \Leftrightarrow \frac{d \log p_t(\mathbf{x}(t))}{dt} = -\operatorname{tr}\left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right)$$

- Likelihood-based CNF training requires ODE solves and Jacobian trace evaluations.
- The adjoint method computes gradients through the ODE solution; repeated solves can be costly.

Chen R. T. Q. et al. Neural Ordinary Differential Equations, 2018

Flow Matching

KFP Theorem (Continuity Equation)

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}\left(\mathbf{v}(\mathbf{x}, t) p_t(\mathbf{x})\right) \Leftrightarrow \frac{d \log p_t(\mathbf{x}(t))}{dt} = -\operatorname{tr}\left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right)$$

- Knowing the vector field $\mathbf{v}(\mathbf{x}, t)$, the KFP (or continuity) equation allows us to compute the density $p_t(\mathbf{x})$.
- Flow matching fits velocities by regression, without model ODE solves or trace evaluations during training.

Flow Matching

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \left\| \mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 \to \min_{\boldsymbol{\theta}}$$

- Approximate the true vector field $\mathbf{v}(\mathbf{x}, t)$ using $\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t)$.
- Use the learned flow $\boldsymbol{\psi}_t$ for deterministic sampling: $\mathbf{x}_1 = \boldsymbol{\psi}_1(\mathbf{x}_0)$.

Flow Matching: Training and Sampling

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})}\left\|\mathbf{v}(\mathbf{x},t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2\to\min_{\boldsymbol{\theta}}$$

$$\mathbf{x}_0\sim p_0(\mathbf{x})=p(\mathbf{x}),\quad \mathbf{x}_1\sim p_1(\mathbf{x})=p_{\text{data}}(\mathbf{x})$$

Training

1. Sample $t\sim U[0,1]$, $\mathbf{x}_t\sim p_t(\mathbf{x})$.
2. Compute loss $\mathcal{L}=\left\|\mathbf{v}(\mathbf{x}_t,t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}_t,t)\right\|^2$.

Sampling

1. Sample $\mathbf{x}_0\sim\mathcal{N}(0,\mathbf{I})$.
2. Solve the ODE to obtain $\mathbf{x}_1$:

$$\mathbf{x}_1=\boldsymbol{\psi}_1(\mathbf{x}_0)=\texttt{ODESolve}_v(\mathbf{x}_0,\boldsymbol{\theta},t_0=0,t_1=1).$$

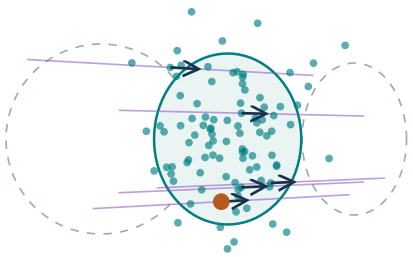
Problem: The true vector field $\mathbf{v}(\mathbf{x},t)$ is **unknown**.

Many Velocity Fields Connect the Same Endpoints

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})}\left\|\mathbf{v}(\mathbf{x},t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2\to\min_{\boldsymbol{\theta}}$$

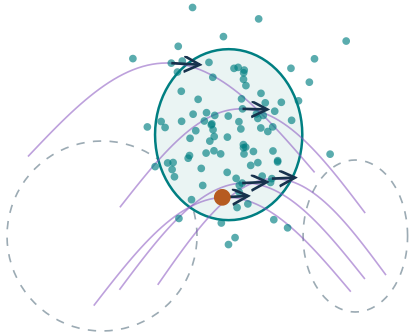
All three flows at t = 0.5. Dashed contours mark the common endpoint densities.

Direct



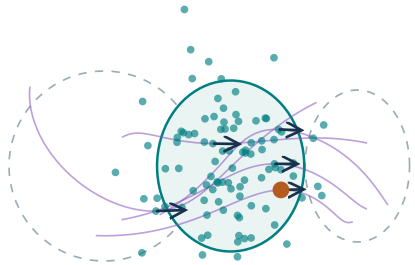
p_0 · noise p_1 · data

Detour



p_0 · noise p_1 · data

Rotation



p_0 · noise p_1 · data

Many velocity fields connect the same endpoints. Choose a path with a tractable training objective.

Intermediate densities may differ. Purple: trajectories; arrows: velocities; orange: one shared particle.

Lipman Y., et al. Flow Matching for Generative Modeling, 2022
Original illustration: A Visual Dive into Conditional Flow Matching

Outline

01 Denoiser Parameterization and Sampling

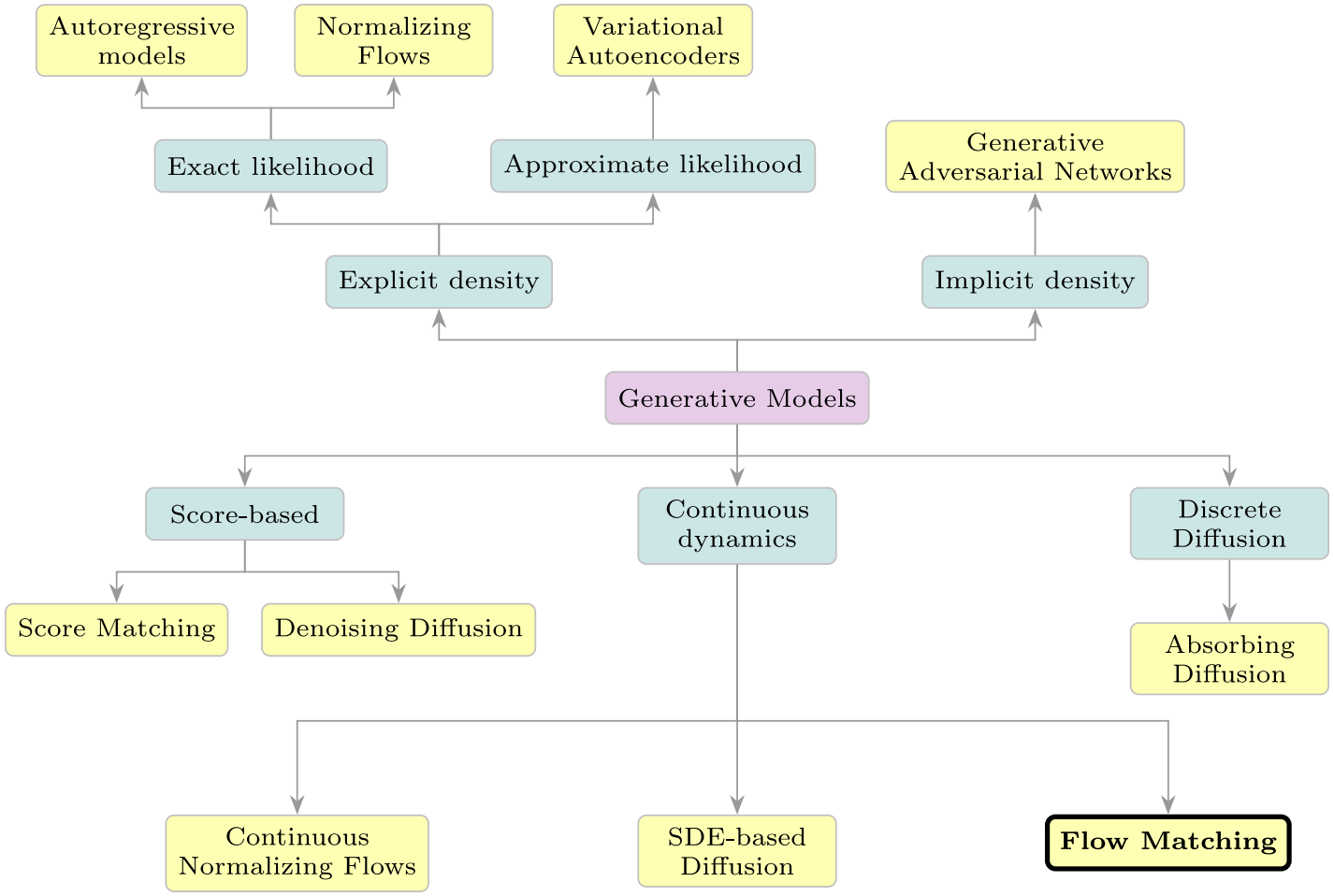
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Generative Models Taxonomy



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From Marginal to Conditional Probability Paths

To build a tractable training target, introduce a latent variable $\mathbf{z}$ and choose conditional paths whose mixture is the desired path:

$$p_t(\mathbf{x}) = \int p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z}$$

Here, $p_t(\mathbf{x}|\mathbf{z})$ is a **conditional probability path**.

The conditional probability path $p_t(\mathbf{x}|\mathbf{z})$ satisfies the KFP theorem:

$$\frac{\partial p_t(\mathbf{x}|\mathbf{z})}{\partial t} = -\operatorname{div}\left(\mathbf{v}(\mathbf{x}, \mathbf{z}, t)p_t(\mathbf{x}|\mathbf{z})\right),$$

where $\mathbf{v}(\mathbf{x}, \mathbf{z}, t)$ is a **conditional vector field**:

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, t) \quad \Rightarrow \quad \frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, \mathbf{z}, t)$$

Conditional Vector Field

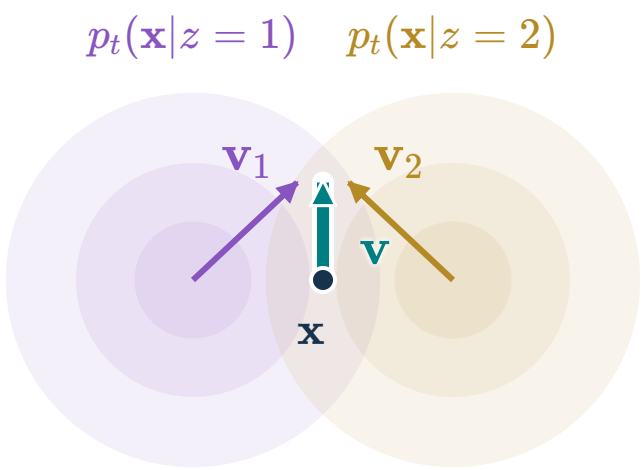
$$\frac{\partial p_t(\mathbf{x}|\mathbf{z})}{\partial t} = -\operatorname{div}\left(\mathbf{v}(\mathbf{x},\mathbf{z},t)p_t(\mathbf{x}|\mathbf{z})\right),$$

What's the relationship between $\mathbf{v}(\mathbf{x},t)$ and $\mathbf{v}(\mathbf{x},\mathbf{z},t)$?

Theorem

The following vector field generates the probability path $p_t(\mathbf{x})$:

$$\mathbf{v}(\mathbf{x},t) = \mathbb{E}_{p_t(\mathbf{z}|\mathbf{x})}\mathbf{v}(\mathbf{x},\mathbf{z},t) = \int \mathbf{v}(\mathbf{x},\mathbf{z},t)\frac{p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{p_t(\mathbf{x})}d\mathbf{z}$$



Each cloud translates with a constant velocity.

Equal priors, fixed time: which cloud is more likely at $\mathbf{x}$?

$p_t(z=1|\mathbf{x})$ = 50% $p_t(z=2|\mathbf{x})$ = 50%

$$\mathbf{v} = 0.50\,\mathbf{v}_1 + 0.50\,\mathbf{v}_2$$

> Move $\mathbf{x}$ → change the posterior → change the velocity.

Conditional Vector Field

Proof

$$
\begin{aligned}
\frac{\partial p_t(\mathbf{x})}{\partial t} &= \frac{\partial}{\partial t} \int p_t(\mathbf{x}|\mathbf{z}) p(\mathbf{z}) d\mathbf{z} \\
&= \int \left(\frac{\partial p_t(\mathbf{x}|\mathbf{z})}{\partial t} \right) p(\mathbf{z}) d\mathbf{z} \\
&= \int \left(-\operatorname{div}\left(\mathbf{v}(\mathbf{x}, \mathbf{z}, t) p_t(\mathbf{x}|\mathbf{z}) \right) \right) p(\mathbf{z}) d\mathbf{z} \\
&= -\operatorname{div}\left(\int \mathbf{v}(\mathbf{x}, \mathbf{z}, t) p_t(\mathbf{x}|\mathbf{z}) p(\mathbf{z}) d\mathbf{z} \right) \\
&= -\operatorname{div}\left(\mathbf{v}(\mathbf{x}, t) p_t(\mathbf{x}) \right)
\end{aligned}
$$

Flow Matching

Flow Matching (FM)

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})}\left\|\mathbf{v}(\mathbf{x},t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2\to\min_{\boldsymbol{\theta}}$$

Conditional Flow Matching (CFM)

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})}\left\|\mathbf{v}(\mathbf{x},\mathbf{z},t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2\to\min_{\boldsymbol{\theta}}$$

Theorem

If $p_t(\mathbf{x})>0$ for all $\mathbf{x}\in\mathbb{R}^m$ and $0\leq t<1$, the FM and CFM objectives differ by a constant independent of $\boldsymbol{\theta}$, so they have the same minimizers.

Proof

This can be proved in a similar way as in the denoising score matching theorem.

Conditional Flow Matching

Theorem

$$\arg\min_{\boldsymbol{\theta}} \mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})} \left\|\mathbf{v}(\mathbf{x},t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2$$

$$= \arg\min_{\boldsymbol{\theta}} \mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})} \left\|\mathbf{v}(\mathbf{x},\mathbf{z},t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2$$

Proof

$$\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})} \left\|\mathbf{v}(\mathbf{x},t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2$$

$$= \mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})} \left[\left\|\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2 - 2\mathbf{v}_{\boldsymbol{\theta}}^{\top}(\mathbf{x},t)\mathbf{v}(\mathbf{x},t)\right] + \text{const}$$

Here const is independent of $\boldsymbol{\theta}$.

Conditional Flow Matching

Proof (continued)

$$
\begin{aligned}
&\mathbb{E}_{p_t(\mathbf{x})}\left[\mathbf{v}_{\boldsymbol{\theta}}^{\top}(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, t)\right] \\
&\quad= \int p_t(\mathbf{x})\left[\mathbf{v}_{\boldsymbol{\theta}}^{\top}(\mathbf{x}, t)\left(\int p_t(\mathbf{z}|\mathbf{x})\mathbf{v}(\mathbf{x}, \mathbf{z}, t)d\mathbf{z}\right)\right]d\mathbf{x} \\
&\quad= \int\int p_t(\mathbf{x})p_t(\mathbf{z}|\mathbf{x})\left[\mathbf{v}_{\boldsymbol{\theta}}^{\top}(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, \mathbf{z}, t)\right]d\mathbf{z}d\mathbf{x} \\
&\quad= \mathbb{E}_{p(\mathbf{z})}\mathbb{E}_{p_t(\mathbf{x}|\mathbf{z})}\left[\mathbf{v}_{\boldsymbol{\theta}}^{\top}(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, \mathbf{z}, t)\right]
\end{aligned}
$$

Conditional Flow Matching: Training and Sampling

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})}\left\|\mathbf{v}(\mathbf{x},\mathbf{z},t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2\to\min_{\boldsymbol{\theta}}$$

$$\mathbb{E}_{p(\mathbf{z})}p_0(\mathbf{x}|\mathbf{z})=\mathcal{N}(0,\mathbf{I});\quad \mathbb{E}_{p(\mathbf{z})}p_1(\mathbf{x}|\mathbf{z})=p_{\text{data}}(\mathbf{x}).$$

Training

1. Sample $t\sim U[0,1]$, $\mathbf{z}\sim p(\mathbf{z})$, $\mathbf{x}_t\sim p_t(\mathbf{x}|\mathbf{z})$.
2. Compute loss $\mathcal{L}=\left\|\mathbf{v}(\mathbf{x}_t,\mathbf{z},t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}_t,t)\right\|^2$.

Sampling

1. Sample $\mathbf{x}_0\sim\mathcal{N}(0,\mathbf{I})$.
2. Solve the ODE to obtain $\mathbf{x}_1$:

$$\mathbf{x}_1=\boldsymbol{\psi}_1(\mathbf{x}_0)=\texttt{ODESolve}_v(\mathbf{x}_0,\boldsymbol{\theta},t_0=0,t_1=1).$$

Marginal-to-Conditional: Scores vs Velocities

Denoising Score Matching

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})}\left\|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma|\mathbf{x})\right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

Optimal predictor over all functions:

$$\mathbf{s}^*_\sigma(\mathbf{x}_\sigma) = \mathbb{E}_{q(\mathbf{x}|\mathbf{x}_\sigma)}\left[\nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma|\mathbf{x})\right] = \nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma)$$

Conditional Flow Matching

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})}\left\|\mathbf{v}(\mathbf{x},\mathbf{z},t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2 \to \min_{\boldsymbol{\theta}}$$

Optimal predictor over all functions: $\mathbf{v}^*(\mathbf{x},t) = \mathbb{E}_{p_t(\mathbf{z}|\mathbf{x})}\mathbf{v}(\mathbf{x},\mathbf{z},t) = \mathbf{v}(\mathbf{x},t)$.

In both cases, the optimal MSE predictor is the posterior mean of the **conditional** target, which equals the intractable **marginal** quantity (score or velocity).

Outline

01 Denoiser Parameterization and Sampling

02 Flow Matching (FM)

03 Conditional Flow Matching (CFM)

CFM Objective
Conditional Probability Paths

04 One-Sided Conditioning

Conditional Flow Matching: Continuity Equations

CFM gives us a training objective. We still need to choose conditional paths and their target velocities.

Continuity Equations

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}\left(\mathbf{v}(\mathbf{x}, t) \cdot p_t(\mathbf{x})\right)$$
$$\frac{\partial p_t(\mathbf{x}|\mathbf{z})}{\partial t} = -\operatorname{div}\left(\mathbf{v}(\mathbf{x}, \mathbf{z}, t) \cdot p_t(\mathbf{x}|\mathbf{z})\right)$$

- $\mathbf{v}(\mathbf{x}, t)$ (under regularity conditions) and initial condition $p_0(\mathbf{x})$ uniquely determine $p_t(\mathbf{x})$.
- Given a field $\mathbf{v}$ generating p_t, adding $\mathbf{w}$ with $\operatorname{div}(p_t \mathbf{w}) = 0$ preserves the same continuity equation.
- We need a velocity field that satisfies the continuity equation and defines a well-posed flow.

Conditional Flow Matching: Paths and Vector Fields

Probability Paths

$$p_t(\mathbf{x}) = \int p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z}$$

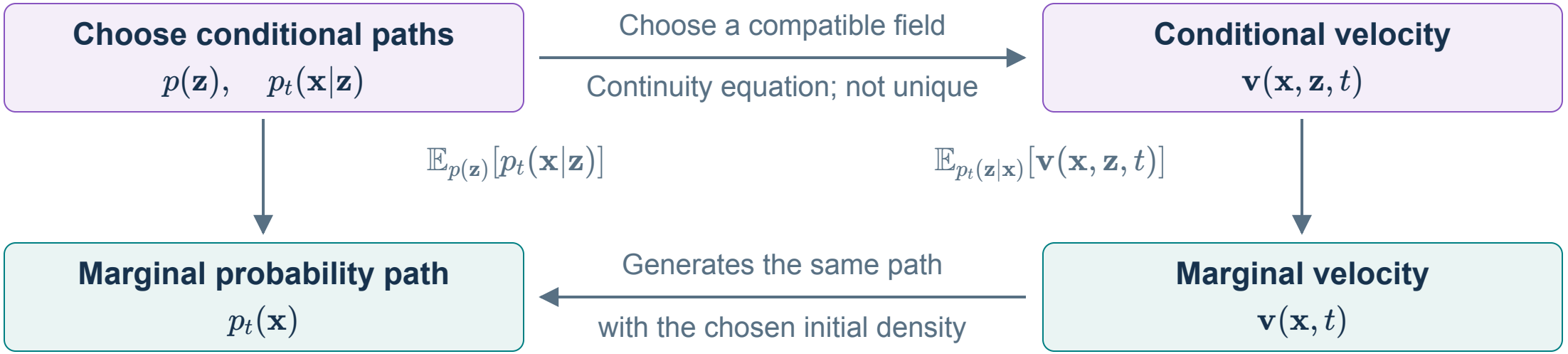
- $p_t(\mathbf{x}|\mathbf{z})$ and $p(\mathbf{z})$ uniquely determine $p_t(\mathbf{x})$.
- The reverse is not true: many different conditional families integrate to the same marginal.

Vector fields

$$\mathbf{v}(\mathbf{x}, t) = \mathbb{E}_{p_t(\mathbf{z}|\mathbf{x})}\mathbf{v}(\mathbf{x}, \mathbf{z}, t) = \int \mathbf{v}(\mathbf{x}, \mathbf{z}, t)\frac{p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{p_t(\mathbf{x})}d\mathbf{z}$$

- $p(\mathbf{z})$, $p_t(\mathbf{x}|\mathbf{z})$ and $\mathbf{v}(\mathbf{x}, \mathbf{z}, t)$ uniquely determine $\mathbf{v}(\mathbf{x}, t)$.
- Choosing conditional paths and compatible velocities constructs both marginal quantities.

From Marginal to Conditional Paths



Open Questions

- **Q1** How should we choose the convenient conditioning latent variable $\mathbf{z}$?
- **Q2** How should we choose $p_t(\mathbf{x}|\mathbf{z})$ so that its marginal satisfies the endpoint constraints?

$$\mathbb{E}_{p(\mathbf{z})}p_0(\mathbf{x}|\mathbf{z}) = \mathcal{N}(0,\mathbf{I});\quad \mathbb{E}_{p(\mathbf{z})}p_1(\mathbf{x}|\mathbf{z}) = p_{\text{data}}(\mathbf{x}).$$

Diagram adapted from A Visual Dive into Conditional Flow Matching

Gaussian Conditional Probability Paths [Q2]

On the interval considered, assume $\boldsymbol{\mu}_t(\mathbf{z})$ and $\boldsymbol{\sigma}_t(\mathbf{z})$ are differentiable in t, with $\boldsymbol{\sigma}_t(\mathbf{z}) > 0$ componentwise.

$$p_t(\mathbf{x}|\mathbf{z}) = \mathcal{N}\left(\boldsymbol{\mu}_t(\mathbf{z}), \boldsymbol{\sigma}_t^2(\mathbf{z})\right)$$

The concrete $\boldsymbol{\mu}_t(\mathbf{z})$, $\boldsymbol{\sigma}_t(\mathbf{z})$ (which satisfy the boundary constraints) will be fixed once we choose $\mathbf{z}$ [Q1].

Gaussian flow

- We need a tractable velocity target for this Gaussian path.
- Choose the affine flow: keep the same Gaussian noise along each trajectory and vary its mean and scale.
- Use this fixed-noise construction:

$$\mathbf{x}_t = \boldsymbol{\psi}_t(\mathbf{x}_0, \mathbf{z}) = \boldsymbol{\mu}_t(\mathbf{z}) + \boldsymbol{\sigma}_t(\mathbf{z}) \odot \boldsymbol{\epsilon},$$

where $\boldsymbol{\epsilon}$ is fixed by the condition: $\mathbf{x}_0 = \boldsymbol{\mu}_0(\mathbf{z}) + \boldsymbol{\sigma}_0(\mathbf{z}) \odot \boldsymbol{\epsilon}$.

Gaussian Conditional Vector Field

Gaussian Conditional Probability Path

$$p_t(\mathbf{x}|\mathbf{z}) = \mathcal{N}\left(\boldsymbol{\mu}_t(\mathbf{z}), \boldsymbol{\sigma}_t^2(\mathbf{z})\right);$$
$$\mathbf{x}_t = \boldsymbol{\psi}_t(\mathbf{x}_0, \mathbf{z}) = \boldsymbol{\mu}_t(\mathbf{z}) + \boldsymbol{\sigma}_t(\mathbf{z}) \odot \boldsymbol{\epsilon}$$

Differentiate the chosen affine flow in t, keeping $\boldsymbol{\epsilon}$ fixed. Assume differentiable $\boldsymbol{\mu}_t, \boldsymbol{\sigma}_t$ and componentwise $\boldsymbol{\sigma}_t > 0$.

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, \mathbf{z}, t); \quad \boldsymbol{\epsilon} = \frac{1}{\boldsymbol{\sigma}_t(\mathbf{z})} \odot (\mathbf{x}_t - \boldsymbol{\mu}_t(\mathbf{z}))$$

$$\frac{d\mathbf{x}_t}{dt} = \boldsymbol{\mu}_t'(\mathbf{z}) + \boldsymbol{\sigma}_t'(\mathbf{z}) \odot \boldsymbol{\epsilon}$$

$$\boxed{\mathbf{v}(\mathbf{x}_t, \mathbf{z}, t) = \boldsymbol{\psi}_t'\left(\boldsymbol{\psi}_t(\mathbf{x}_t, \mathbf{z})^{-1}, \mathbf{z}\right) = \boldsymbol{\mu}_t'(\mathbf{z}) + \frac{\boldsymbol{\sigma}_t'(\mathbf{z})}{\boldsymbol{\sigma}_t(\mathbf{z})} \odot (\mathbf{x}_t - \boldsymbol{\mu}_t(\mathbf{z}))}$$

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Endpoint Conditioning

Conditional Flow Matching

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})}\left\|\mathbf{v}(\mathbf{x},\mathbf{z},t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2\to\min_{\boldsymbol{\theta}}$$

Let's define our latent variable $\mathbf{z}$.

Conditioning Latent Variable [Q1]

Let us choose $\mathbf{z}=\mathbf{x}_1$. Then $p(\mathbf{z})=p_1(\mathbf{x}_1)$.

$$p_t(\mathbf{x})=\int p_t(\mathbf{x}|\mathbf{x}_1)p_1(\mathbf{x}_1)d\mathbf{x}_1$$

We need to ensure the boundary constraints:

$$\begin{cases}\mathbb{E}_{p(\mathbf{z})}p_0(\mathbf{x}|\mathbf{z})=\mathcal{N}(0,\mathbf{I})\\ \mathbb{E}_{p(\mathbf{z})}p_1(\mathbf{x}|\mathbf{z})=p_{\text{data}}(\mathbf{x})\end{cases}\Rightarrow\quad\begin{cases}p_0(\mathbf{x}|\mathbf{x}_1)=\mathcal{N}(0,\mathbf{I});\\ p_1(\mathbf{x}|\mathbf{x}_1)=\delta(\mathbf{x}-\mathbf{x}_1)\end{cases}$$

One-Sided Conditioning

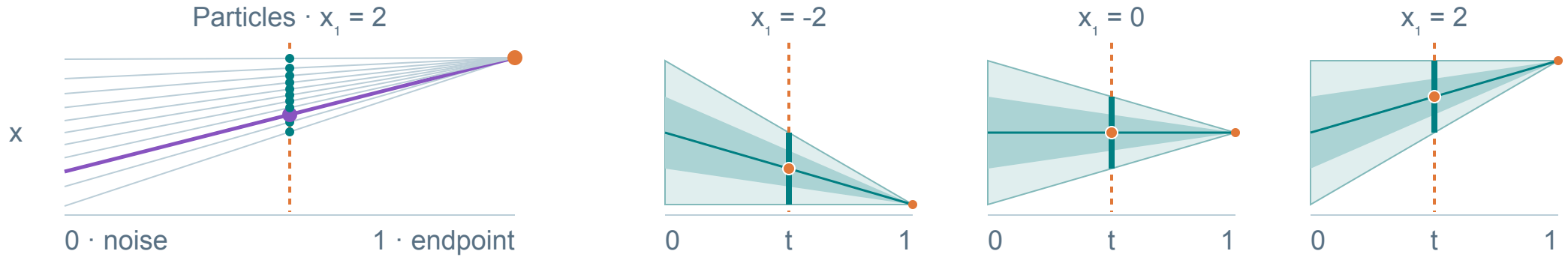
$$p_0(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}(0, \mathbf{I}); \quad p_1(\mathbf{x}|\mathbf{x}_1) = \delta(\mathbf{x} - \mathbf{x}_1)$$

Gaussian Conditional Probability Path

$$p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}\left(\boldsymbol{\mu}_t(\mathbf{x}_1), \boldsymbol{\sigma}_t^2(\mathbf{x}_1)\right);$$
$$\mathbf{x}_t = \boldsymbol{\mu}_t(\mathbf{x}_1) + \boldsymbol{\sigma}_t(\mathbf{x}_1) \odot \boldsymbol{\epsilon}$$

Let's consider straight conditional paths:

t = 0.5 · fixed particles; shaded bands: mean ± σ / ± 2σ.



$$\begin{cases} \boldsymbol{\mu}_t(\mathbf{x}_1) = t\mathbf{x}_1 \\ \boldsymbol{\sigma}_t(\mathbf{x}_1) = 1 - t \end{cases} \Rightarrow \begin{cases} p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}\left(t\mathbf{x}_1, (1-t)^2 \cdot \mathbf{I}\right) \\ \mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0 \end{cases}$$

One-Sided Conditioning: Conditional Vector Field

$$p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}\left(t\mathbf{x}_1, (1-t)^2\mathbf{I}\right); \quad \mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0$$

Conditional Vector Field

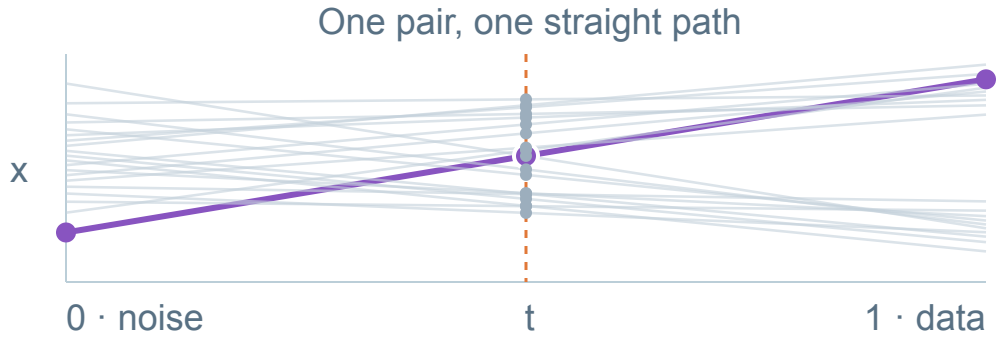
$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t) = \boldsymbol{\mu}_t'(\mathbf{x}_1) + \frac{\boldsymbol{\sigma}_t'(\mathbf{x}_1)}{\boldsymbol{\sigma}_t(\mathbf{x}_1)} \odot (\mathbf{x}_t - \boldsymbol{\mu}_t(\mathbf{x}_1))$$

$$\mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t) = \mathbf{x}_1 - \frac{1}{1-t} \cdot (\mathbf{x}_t - t\mathbf{x}_1) = \frac{\mathbf{x}_1 - \mathbf{x}_t}{1-t}, \qquad t < 1$$

$$= \frac{\mathbf{x}_1 - t\mathbf{x}_1 - (1-t)\mathbf{x}_0}{1-t}$$

$$= \mathbf{x}_1 - \mathbf{x}_0$$

At $t = 1$, the quotient is undefined. Use the simplified training target $\mathbf{x}_1 - \mathbf{x}_0$, which contains no division.



$$x_0 = -1.96, \qquad x_1 = 2.07$$

$$v = x_1 - x_0 = 4.03$$

Move along the path: the velocity stays constant.

t = 0.5 · A fixed pair has a constant target velocity.

One-Sided Conditioning: CFM Objective

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})}\left\|\mathbf{v}(\mathbf{x},\mathbf{z},t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2$$

$$=\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{x}_1\sim p_{\text{data}}(\mathbf{x})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{x}_1)}\left\|\frac{\mathbf{x}_1-\mathbf{x}}{1-t}-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2$$

$$=\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{x}_1\sim p_{\text{data}}(\mathbf{x})}\mathbb{E}_{\mathbf{x}_0\sim\mathcal{N}(0,\mathbf{I})}\left\|(\mathbf{x}_1-\mathbf{x}_0)-\mathbf{v}_{\boldsymbol{\theta}}\left(t\mathbf{x}_1+(1-t)\mathbf{x}_0,t\right)\right\|^2$$

Rectified Flow

Regressing on $\mathbf{x}_1-\mathbf{x}_0$ along $\mathbf{x}_t=(1-t)\mathbf{x}_0+t\mathbf{x}_1$ gives the **rectified flow objective**.

Straight training paths do not imply straight sampling trajectories.

$$\mathbf{v}_{\boldsymbol{\theta}^*}(\mathbf{x},t)=\mathbb{E}\left[\mathbf{x}_1-\mathbf{x}_0\mid\mathbf{x}_t=\mathbf{x}\right].$$



t = 0.5 · Same initial particles; endpoint pairings can differ.

Lipman Y., et al. Flow Matching for Generative Modeling, 2022
Liu X., Gong C., Liu Q. Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow, 2022

Rectified Flow: Training and Sampling

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{x}_1\sim p_{\text{data}}(\mathbf{x})}\mathbb{E}_{\mathbf{x}_0\sim\mathcal{N}(0,\mathbf{I})}\left\|(\mathbf{x}_1-\mathbf{x}_0)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}_t,t)\right\|^2\to\min_{\boldsymbol{\theta}}$$

Training

1. Sample $\mathbf{x}_1\sim p_{\text{data}}(\mathbf{x})$, $\mathbf{x}_0\sim\mathcal{N}(0,\mathbf{I})$, $t\sim U[0,1]$.
2. Compute noisy image $\mathbf{x}_t=t\mathbf{x}_1+(1-t)\mathbf{x}_0$.
3. Compute loss $\mathcal{L}=\left\|(\mathbf{x}_1-\mathbf{x}_0)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}_t,t)\right\|^2$.

Sampling

1. Sample $\mathbf{x}_0\sim\mathcal{N}(0,\mathbf{I})$.
2. Solve the ODE to obtain $\mathbf{x}_1$:

$$\mathbf{x}_1=\boldsymbol{\psi}_1(\mathbf{x}_0)=\texttt{ODESolve}_v(\mathbf{x}_0,\boldsymbol{\theta},t_0=0,t_1=1).$$

Lipman Y., et al. Flow Matching for Generative Modeling, 2022
Liu X., Gong C., Liu Q. Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow, 2022

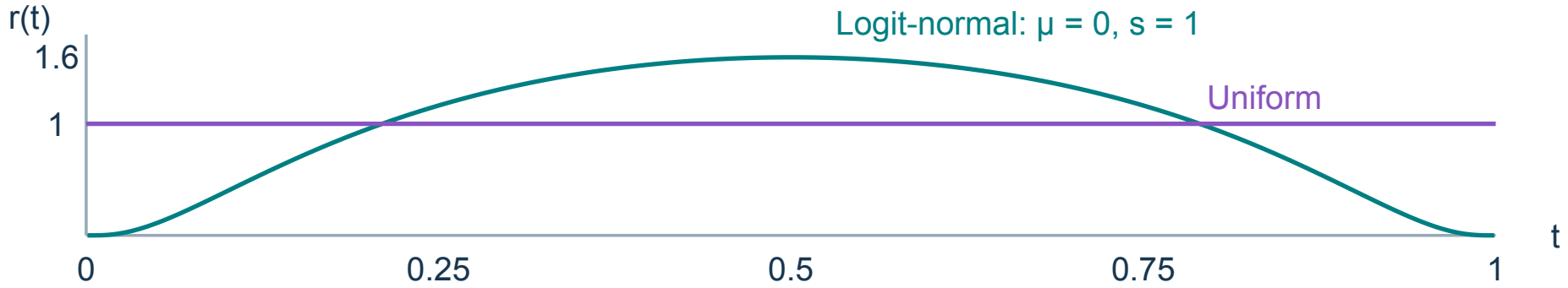
Rectified Flow: Sampling Training Times

For $\mathbf{x}_t = (1-t)\mathbf{x}_0 + t\mathbf{x}_1$, let $\ell_{\boldsymbol{\theta}}(t) = \mathbb{E}\|\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) - (\mathbf{x}_1 - \mathbf{x}_0)\|^2$.

$$\mathbb{E}_{t\sim r}\ell_{\boldsymbol{\theta}}(t) = \mathbb{E}_{t\sim U[0,1]}\left[r(t)\ell_{\boldsymbol{\theta}}(t)\right].$$

Without importance correction, more frequent times receive more weight.

$$u \sim \mathcal{N}(\mu, s^2), \qquad t = \frac{1}{1+e^{-u}} \quad \Rightarrow \quad \log \mathrm{SNR}(t) = 2\log\frac{t}{1-t} = 2u.$$



Logit-normal time means Gaussian log-SNR. SD3 explores this reweighting. The probability path and the target $\mathbf{x}_1 - \mathbf{x}_0$ stay fixed.

Classifier-Free Guidance for Velocity Fields

Recall diffusion CFG for a label/prompt $\mathbf{y}$ (distinct from the CFM latent $\mathbf{z}$):

$$\mathbf{s}^{\gamma}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y}) = \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \varnothing) + \gamma \left[\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y}) - \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \varnothing)\right].$$

$$\mathbf{v}^{\gamma}_{\boldsymbol{\theta}}(\mathbf{x}, t, \mathbf{y}) = \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t, \varnothing) + \gamma \left[\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t, \mathbf{y}) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t, \varnothing)\right].$$

$\gamma = 0$: unconditional; $\gamma = 1$: conditional; $\gamma > 1$: extrapolation.

Training

1. Sample paired data $(\mathbf{x}_1, \mathbf{y}) \sim p_{\text{data}}$ and form the rectified-flow sample $\mathbf{x}_t$ as before.
2. Set $\tilde{\mathbf{y}} = \varnothing$ with fixed probability; otherwise set $\tilde{\mathbf{y}} = \mathbf{y}$.
3. Minimize the same velocity loss:

$$\mathcal{L} = \left\| \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}_t, t, \tilde{\mathbf{y}}) - (\mathbf{x}_1 - \mathbf{x}_0) \right\|^2.$$

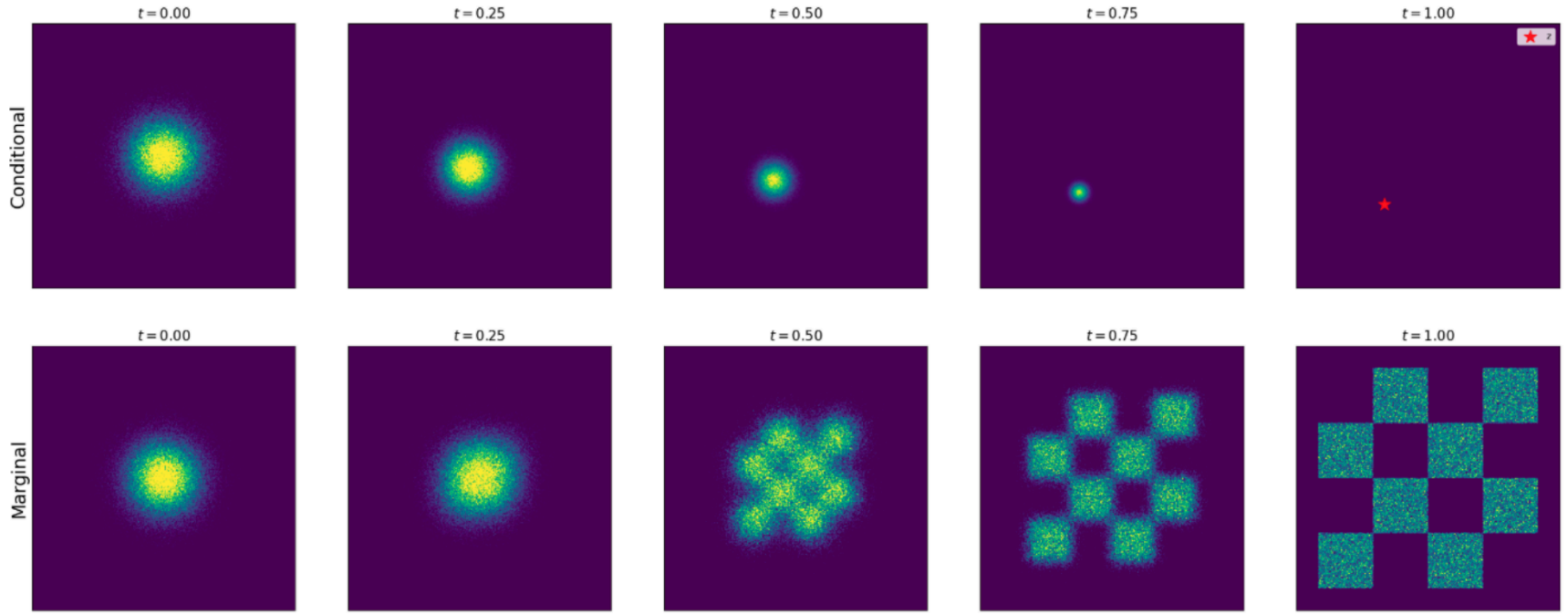
Sampling

1. Sample $\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})$ and choose $\mathbf{y}, \gamma$.
2. Evaluate both predictions to form $\mathbf{v}^{\gamma}_{\boldsymbol{\theta}}$ at each solver evaluation.
3. Solve the guided ODE from $t = 0$ to $t = 1$:

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}^{\gamma}_{\boldsymbol{\theta}}(\mathbf{x}_t, t, \mathbf{y}).$$

Ho J., Salimans T. Classifier-Free Diffusion Guidance, 2022
Esser P., et al. Scaling Rectified Flow Transformers for High-Resolution Image Synthesis, 2024

Conditional vs. Marginal Paths



- One-sided conditioning gives us the way to construct a generative model.
- Now we extend it to image-to-image formulation (mapping between two distinct data distributions $p_0(\mathbf{x})$ and $p_1(\mathbf{x})$).

Summary

- A denoiser estimates the clean conditional mean under Gaussian noise. In noise coordinates, deterministic DDIM is an Euler step of the denoiser PF-ODE.
- Many velocity fields connect the same endpoints. FM fits a chosen field by regression, without ODE solves or Jacobian traces during training.
- CFM preserves the FM minimizers. The marginal velocity is the posterior mean $\mathbf{v}(\mathbf{x}, t) = \mathbb{E}_{p_t(\mathbf{z}|\mathbf{x})}\mathbf{v}(\mathbf{x}, \mathbf{z}, t)$.
- Gaussian conditional paths admit a closed-form conditional velocity field.
- Changing the training-time distribution without importance correction reweights the loss; logit-normal time gives Gaussian log-SNR on the straight noise-to-data path.
- For straight paths conditioned on $\mathbf{x}_1$, **rectified flow** regresses on the displacement $\mathbf{x}_1 - \mathbf{x}_0$.
- Condition dropout trains both velocity predictions used in classifier-free guidance.